35KW Amp Head Maximum Power Setup and Calibration

(Click here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	45 mins	Not Applicable

2 Overview

Last Update	03/02/2005

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Power Measurement Kit (Wattmeter Kit)	1	-	2372868	-
Medium Phillips Screwdriver	1	-	-	-
12 in Crescent Wrench	1	-	-	-

3.2 Safety

DANGER

FATAL ELECTRIC SHOCK HAZARD
THE RF AMPLIFIER IS NOT DISABLED
TO PREVENT FATAL ELECTRIC SHOCK, ENSURE THE RF AMPLIFIER IS DISABLED WHILE CONFIGURING THE CALIBRATION TOOLS.

NOTICE

Property Damage! Prevent Coil And Associated Switch Damage, By Removing All Phantoms And Hardware (I.E., Head Coil, Surface Coil...) From The Magnet Bore

NOTICE

Possible equipment damage. Do not operate the RF amplifier with an improper load connected to its output. Doing so may permanently damage the amplifier

NOTICE

Possible equipment damage. Do not turn connect/disconnect Wattmeter Sensor while Meter is ON.

Always turn Meter OFF prior to plugging/unplugging Sensor.

4 Procedure

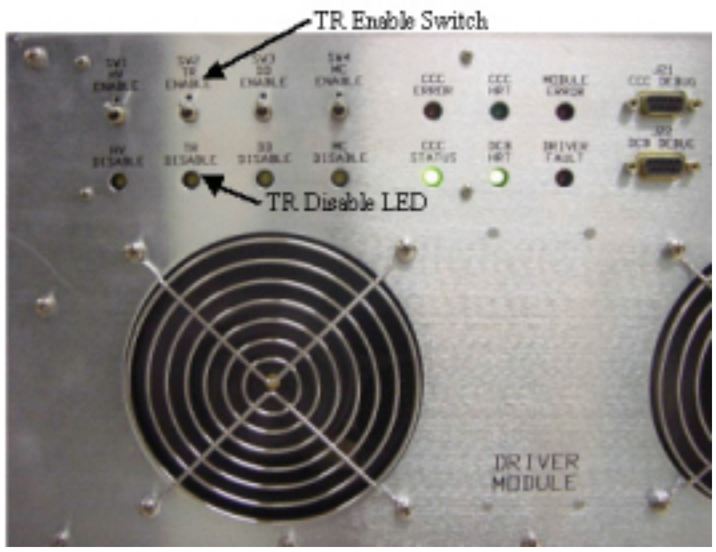
4.1 DISABLING T/R DRIVE ERROR CONDITION (Production Systems Using TR In Driver Module)

1. Verify that the system is idle and all coils have been removed from magnet bore.

2. Remove the front cover of the EXCITE Systems Cabinet.

3. On the front of the Driver Module, disable T/R error reporting by setting the TR switch to the DIS (disable faults) position. (see Illustration 1).

Illustration 1: Driver Module Switch

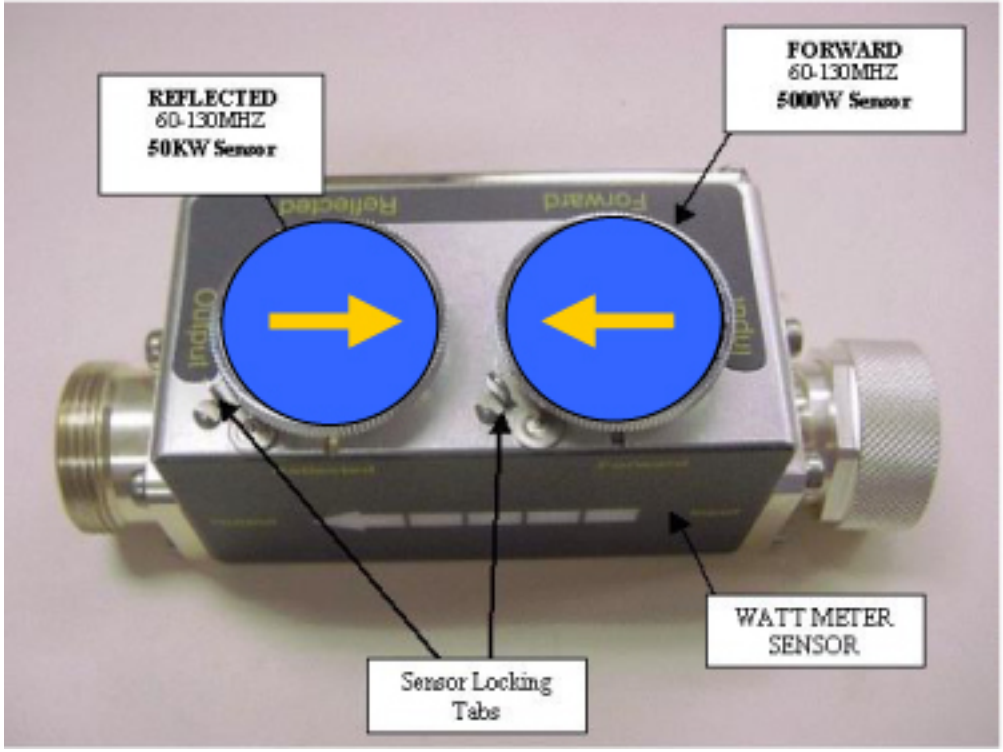


NOTE: **HV**, **DD** and **MC** should remain set to Enable.

4.2 HEAD- MAXIMUM POWER CALIBRATION

1. Insert the 5000W Element into the Forward Sensor Port, (Blue- 4300A374-6). Make sure the arrow on the element points in the direction of the arrow (from Input to Output) on the Sensor. Lock Element in place using the locking tab on the sensor. ((see Illustration 2))

Illustration 2: RF Coupler Setup For Head Mode



2. Insert the 50KW Element into the Reflected Sensor Port. (Blue- 4300A374-3). Make sure the arrow on the element points in the opposite direction of the arrow (from Input to Output) on the Sensor. Lock Element in place using the locking tab on the sensor. ((see Illustration 2)). The reflected port is not used during this procedure. Placing the 50kW element into the reflected port is just a safety measure.

4.3 Wattmeter and Dummy Load Setup

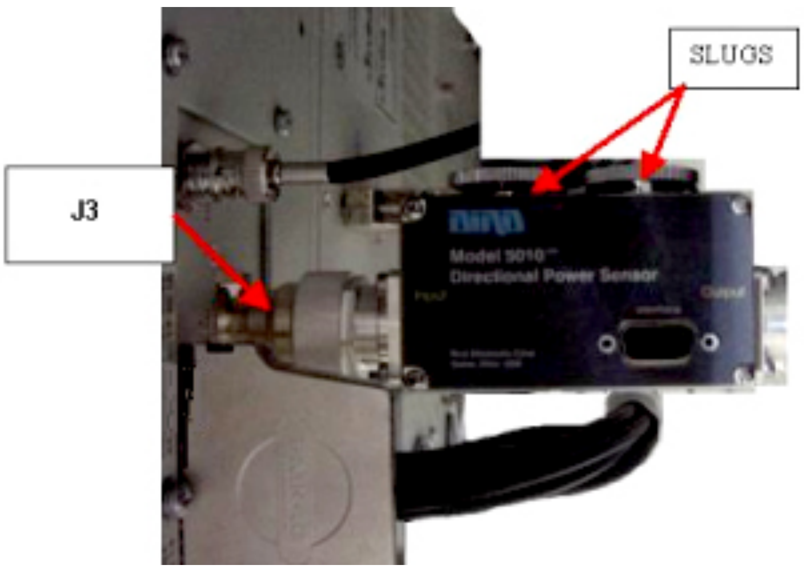
1. Verify scanner is in "idle", turn off RF Enable at the exciter (FP-I/O board) or disconnect J14 (RF Drive input) from the back of the RF amplifier.

2. Disconnect the Head cable J3 "Head Output" from the back of the RF Amplifier.

3. To protect the Body hybrid and amplifier section, disconnect the Body cable J4 "Body Output" from the back of the RF Amplifier as well.

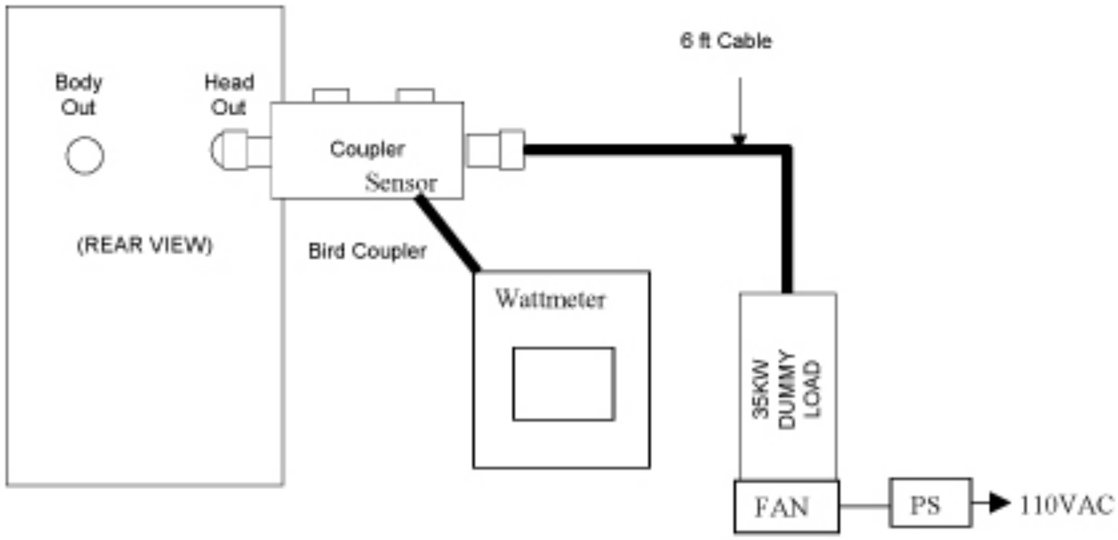
4. Connect the Input of the Coupler to J3 of the RF Amplifier. (Note: Some new RF Kits may require using the HN to 7/16 adapter included in the kit. ((see Illustration 3)).

Illustration 3: Connection Of RF Coupler To Head Output



5. Connect the RF Coupler output to the Dummy Load ((see Illustration 4)).

Illustration 4: Complete Head Output Test Setup



6. Connect one end of the Wattmeter cable to the Wattmeter sensor output. (RS232 is unused). Connect the other end to the coupler.
7. Make sure to plug in the fan for the dummy load. AT NO TIME should you use this dummy load without its fan running.

NOTICE

DO NOT APPLY RF POWER TO THE DUMMY LOAD WITHOUT ITS FAN IN OPERATION. DOING SO WILL RESULT IN CATASTROPHIC DAMAGE TO THE DUMMY LOAD.

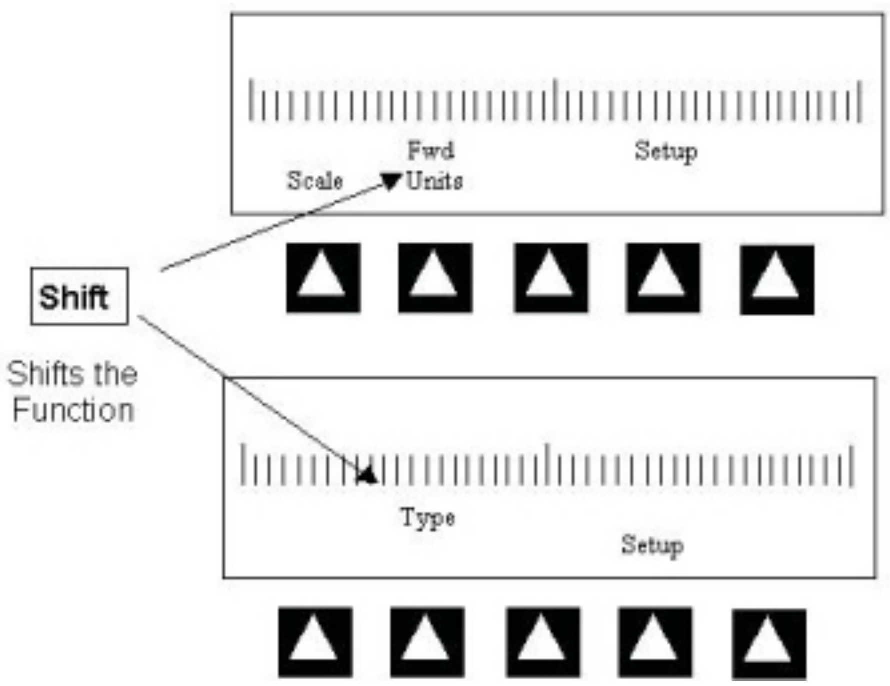
8. Reconnect J14 on the back of the amplifier or switch RF Enable to **ON**.

4.4 Wattmeter introduction

For more information on the Watt Meter, *Watt Meter (2372868) Introduction*.

1. Simply switching it off and then back on resets the Digital Wattmeter. If you get confused press the **On** button twice and start the procedure over.
2. The Arrow Buttons Point to the corresponding item within the LCD Display, directly above the button. ((see Illustration 5)):

Illustration 5: Watt Meter Operation



Pressing the <Shift> Button shifts the function of the <Arrow Key>.

4.5 Setting Up The Wattmeter to Measure Forward Head Maximum Power

1. Press the **ON** button on the Wattmeter. (If coupler and sensor is not correctly attached to the amplifier you will get a “No Sensor” indication)

2. Select the △arrow button under *Scale*.

3. Type: **5000** to set the range.

4. Select <Enter> button

5. Check the setting, by selecting the arrow △ button under *Scale* once again.

6. Press Enter to exit this mode.

7. Select the arrow △ button under Fwd Units make sure the scale reads “W” (or “kW “on some watt meters). If the scale reads dBm, select the arrow △ button under Fwd Units again, to toggle the scale to read “W” (Watts or kW kiloWatts) on the TOP, Forward Power scale. Ignore the BOTTOM, reflected Power scale value. It is not used during Maximum Power Calibration.

8. Select the <Shift> button.

9. Select the arrow △ button under Setup.

10. Verify it reads “43”. If not, type the number “43” followed by the enter button. (Illustration 6 shows what should be displayed at this time).

Illustration 6: WATTMETER DISPLAY SET TO READ PEAK POWER SENSOR

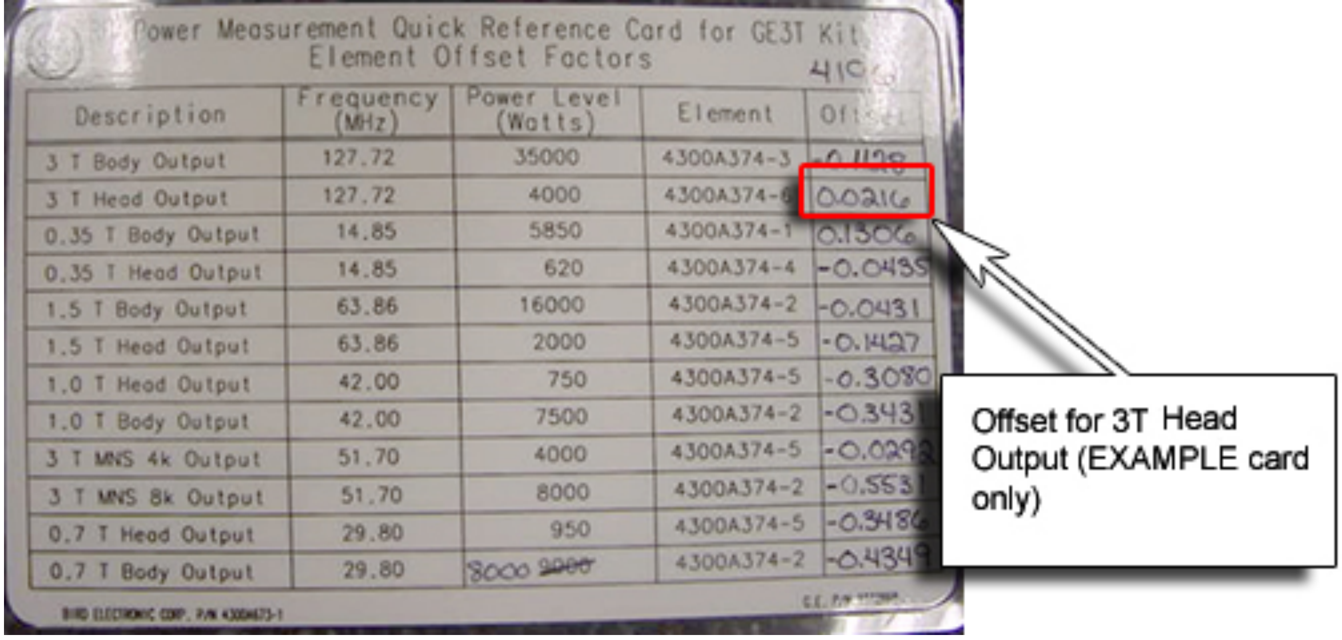
NOTE: This action displays the Sensor Type configured in the Wattmeter. This procedure uses Sensor” Type 43”, which is optimized for peak power measurement.

11. Select <Enter> button.

12. Select the arrow button under Type, change unit of measurement to Peak.

13. Some kits require an offset to ensure accurate measurement. Locate the Element Offset Factors card in the kit and view the offset for 3T Head Output. See Illustration 7.

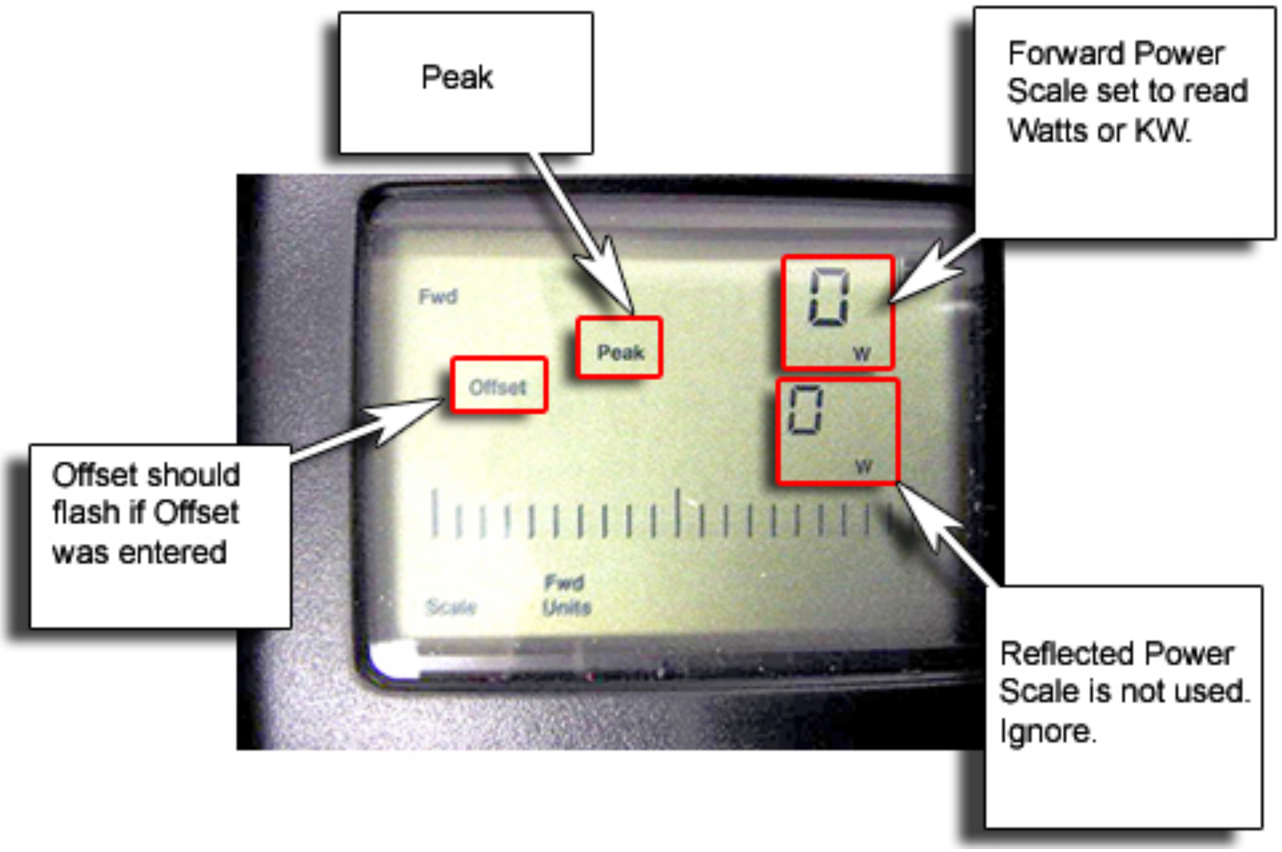
Illustration 7: Element Offset Factor Card


Power Measurement Quick Reference Card for GE3T Kit
Element Offset Factors

Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T Body Output	127.72	35000	4300A374-3	0.1128
3 T Head Output	127.72	4000	4300A374-6	0.0216
0.35 T Body Output	14.85	5850	4300A374-1	0.1306
0.35 T Head Output	14.85	620	4300A374-4	-0.0435
1.5 T Body Output	63.86	16000	4300A374-2	-0.0431
1.5 T Head Output	63.86	2000	4300A374-5	-0.1427
1.0 T Head Output	42.00	750	4300A374-5	-0.3080
1.0 T Body Output	42.00	7500	4300A374-2	-0.3431
3 T MNS 4k Output	51.70	4000	4300A374-5	-0.0292
3 T MNS 8k Output	51.70	8000	4300A374-2	-0.5531
0.7 T Head Output	29.80	950	4300A374-5	-0.3486
0.7 T Body Output	29.80	8000 ~~9000~~	4300A374-2	-0.4349

14. If an offset exists, press the Offset Button on the wattmeter. Enter in the value that was diplayed on the kits offset card and press enter. (Illustration 7 shows what should be displayed at this time).

Illustration 8: Wattmeter Set To Display Peak Power And Watts



15. The meter is now configured to read RF Peak Power, in Watts for the Head Maximum Output calibration.

4.6 Scan

NOTICE

Property Damage! Prevent coil and associated switch damage, by Removing all phantoms and hardware (i.e., Head Coil, surface Coil...) from the magnet bore.

1. Prepare the system to scan in head mode:

   a. Click on [New Pt]

   b. Id: **geservice**<Enter>

c. Name: **`apb cal Head`**<Enter>

d. Weight (Lb): **`300`**<Enter>

e. Set Patient Protocols to [Service]

f. Scroll through the protocol field until you find thee protocol [0.41-3.0T apb call].

g. Select [Series 2, Head]

h. [Accept]

i. Set **Landmark**

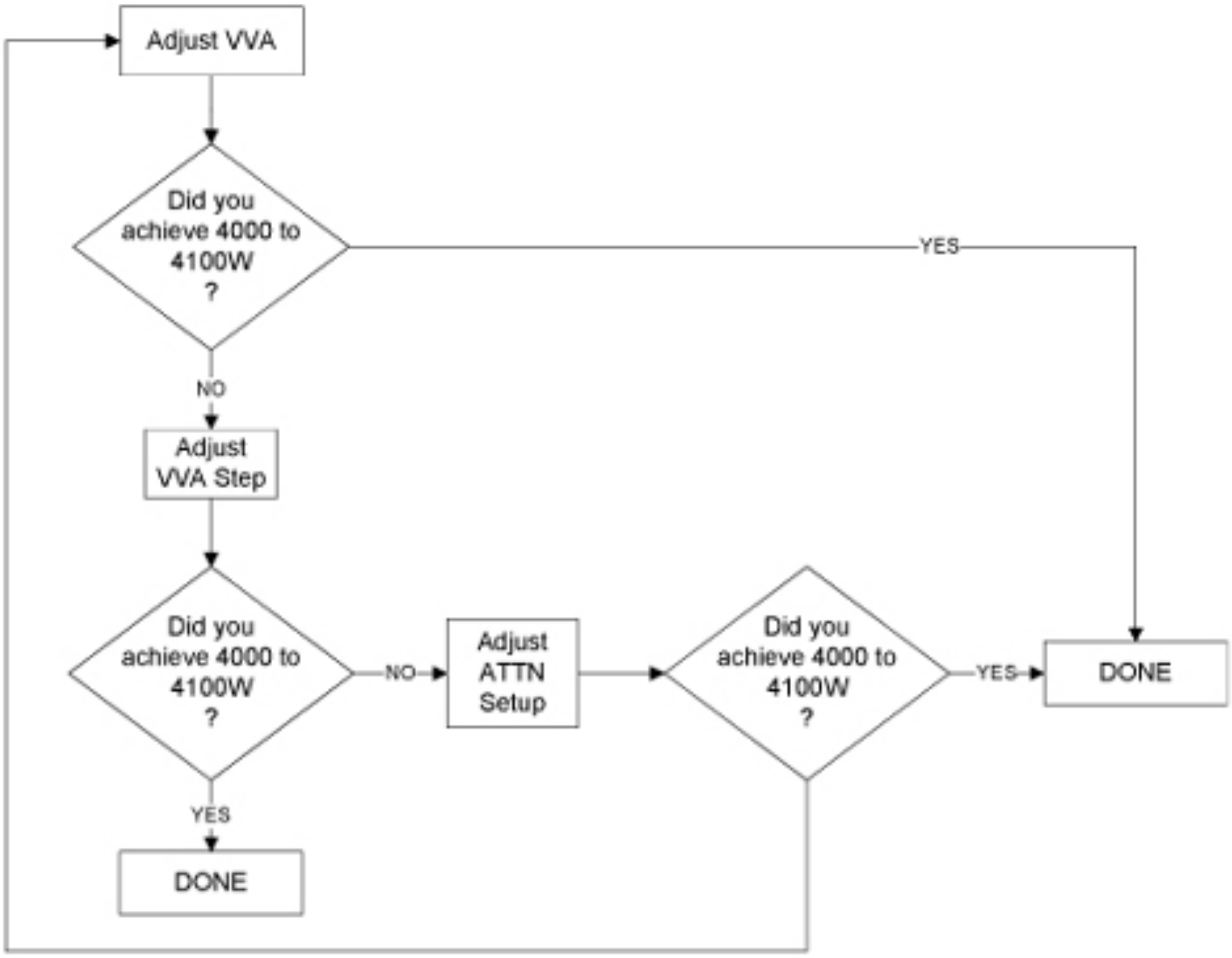
2. [Save Series]
3. Select [Research Options] with the mouse cursor, and right click. Then select [Setup Params]. Set TG to **`180`**[Done].
4. Select [Research Options] with the mouse cursor, and right click, select [Display CVs] From the drop down menu.
5. Set value of CV *calmode* to **`5`**<Enter> (Dual Logamp Waveform). (Make sure to press <Enter> to capture the setting.
6. Select [Accept], then [Download].
7. Go to the amplifier and remove the Keypad cover, located on the front of the amplifier near the display.

Illustration 9: FRONT OF RF AMP CONTROL MODULE

4.7 Adjusting the RF Amplifier Output Power (Head)

1. This section adjusts the Solid State Amplifier gain of the Solid State pre-amplifier (VVA Setup) and if necessary, also adjusts the Attenuation of the IPA stage, (ATTN Setup). The Gain (VVA- Variable Voltage Adjustment) of the Solid-state amplifier is setup first, maximizing its output. The Attenuation of the IPA stage can also be adjusted to “Fine tune” the final value to as close as possible to 4000 Watts, (4000-4100 Watts) .VVA Setup and ATTN Setup interact with each other, and you may find yourself jumping back and forth between these adjustments until the amplifier is properly calibrated.

Illustration 10: Process Flow



2. Select [Manual Prescan].

3. Select [Scan TR].

4. Set the Transmit Gain to **`180`**. Make sure everything is functioning normally and you are getting a reading on the Wattmeter. Allow scanner to pulse at **`TG 180`** for **`6`** minutes. This is due to the drop in gain that occurs over the 6 minutes period.

5. Slowly increase the TG to 200. (If the amplifier trips set the TG at its highest setting). Resume the procedure at VVA SETUP (Solid State amplifier Gain adjust).

4.8 VVA SETUP (Solid State amplifier Gain adjust)

NOTICE

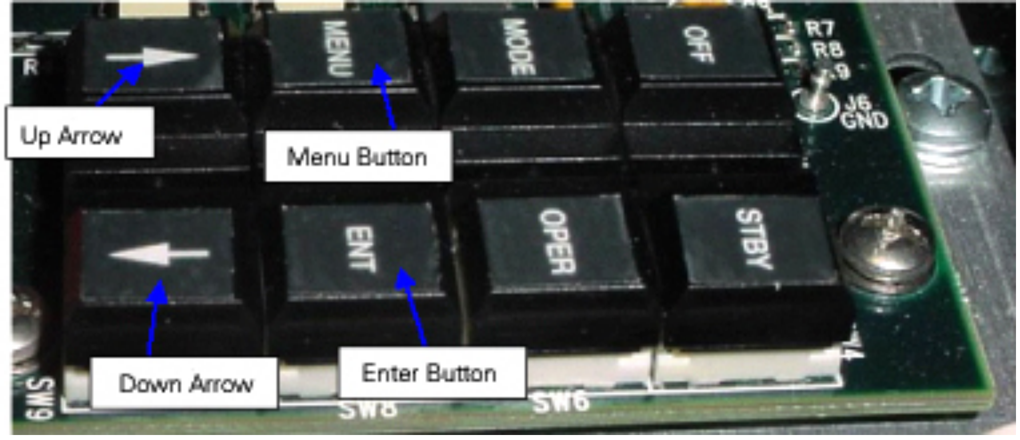
Press Only The Buttons Required By This Procedure. Avoid Pressing Mode Button. It Controls The Output Drive Of The Solid State Amplifier, Switching From Body To Head Mode Of Operation. For information on the key pad, see *Keyboard Operation / Description of IPA and PA Stages*.

1. Refer to Illustration 11. While the system is pulsing RF, Press the <Menu> Button on the Keypad located inside the RF amplifier, until the LCD Display on the front of the amplifier reads the following:

   > ATTN SETUP

   VVA SETUP

Illustration 11: RF SETUP KEYPAD



2. Press <Down Arrow> to select VVA SETUP

   ATTN SETUP

   >VVA SETUP

3. Press <Enter>.

4. Continue to press <Enter> button until you've selected the desired step size. For smaller and finer calibration use smaller incremental step sizes such as 1 or 10. (Step size values can be set to 1,10,20,30,40,50,60,70,80,90,or 100).

5. Once the Step Size is selected, Press Down or up arrow to actually adjust the Gain:

   VVA Step: 10

   >VVA: 3368 (This value will change by 10 counts)

6. Using the Down/Up arrows resume the adjustment of the gain until you have achieved 4000 Watts (Approx. 4000-4100 Watts) on the meter.

7. If you can achieve 4000Watts on the meter display Press the Menu button until you reset the keypad to read Forward and Reflected Power.

   NOTE: If you are still not able to achieve the proper maximum Wattage setting, you may need to adjust the IPA stage Attenuation, see Adjusting ATTN SETUP (PA Attenuation).

8. Proceed to Saving New Calibration Settings Saving New Calibration Settings.

4.9 Adjusting ATTN SETUP (PA Attenuation)

1. This section only needs to be performed if you can't achieve 4000 Watts (Approx. 4000-4100 Watts) with VVA SETUP. (Note: At this time the RF amplifier is still pulsing at TG of 200)

2. Press the <Menu> Button until the LCD Display on the front of the amplifier reads the following:

   >ATTN SETUP

   VVA SETUP

3. Select <Enter>.

   > ATTN: **13** (This value will change when the down/up arrows are pressed)

4. Press the Down or Up arrows while watching the Wattmeter to adjust the attenuation of the RF amplifier PA stage. Slowly attenuate the RF power to as close to 4000 Watts as you can. If you cannot achieve 4000 Watts, adjust the attenuation to a value that is below 4000 Watts.

5. Press the <Menu> button to exit.

6. Press the <Menu> button until amp displays Forward and Reflected power readings.

7. If you did not achieve 4000 Watts, proceed to VVA SETUP (Solid State amplifier Gain adjust). If you achieved 4000W (Approx. 4000-4100 Watts) proceed to Saving New Calibration Settings.

4.10 Saving New Calibration Settings

NOTE: The RF amplifier should still be producing RF at a TG of 200.

1. As a quick check, look at the LCD display on the front of the amplifier. It should be reading 4000 + or - approximately 600 Watts. This will vary site to site and can be used as a confidence indicator as to your actual setting. The LCD display should not be relied on as an accurate measurement device.

2. Stop [Manual Prescan].

3. Press the <Menu> button multiple times until the original screen on the amp LCD reappears.

4. Press the **Off** button. This will STORE the value in NVRAM. Wait 30 seconds and press the “Operate” (Or “standby) button. Wait another 30 seconds for the amplifier to react to the new settings.

5. Reduce the TG to zero, once the RF Output is calibrated to specification. If the Amplifier is setup correctly, the RF amplifier Display, and Wattmeter should be very close to the same value.

6. Test the new Maximum Power value by restarting [Manual Prescan].

7. Select [Scan TR].

8. Slowly increase the TG to **200**. The system should not trip.

9. Take a final look at the Wattmeter display; making sure the desired value is displayed.

10. Reduce TG to Zero.

11. Stop [Manual Presecan]

12. Continue with Finalization Step .

5 Finalization

WARNING

ELECTRICAL/RF BURN HAZARD
RF BURNS AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING
PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RF AMP BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE “IDLE” MESSAGE AND THE RF ENABLE SWITCH IS IN THE "OFF" POSITION

NOTICE

Possible equipment damage. Do not turn connect/disconnect Wattmeter Sensor while Meter is ON. Always turn Meter OFF prior to plugging/unplugging Sensor

1. Verify scanner is in “idle”, turn off RF Enable and disconnect J14 from the back of the RF amplifier.

2. Before proceeding, perform a [Reset TPS]. This is to ensure that the RF Amplifier is in a known state to the system.

3. Perform UPM *(Universal Power Monitor - Head Setup and Calibration)* to ensure the UPM is properly calibrated and functioning properly.

4. Disconnect the RF Watt Sensor from the J3 Head Output Connection

5. Connect the Original Head Output Cable to RF Amplifier, J3 and Body Output able to RF Amplifier J4.

Back to top